Pre Analysis Plan: #Manydaughters

The goal of the project is to verify four different hypotheses related to how having a daughter affects parents' perspective and actions. Concretely, we are set to test four different hypotheses using SOEP data. The hypotheses are:

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).
H2. Having daughters leads to more left-leaning political preferences.
H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

The hypotheses should be tested as causally, and not as a correlation.

Assumptions:
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Hypotheses 1 and 4:

The most important assumption of our analysis is that conditional on being pregnant the gender of the child is random. We presume that selective abortion is rare / nonexistent in Germany. Under this scenario, the gender of the first child is exogenous and variation of this variable can be used to test H1 and H4. Notice that we do not assume that pregnancy is exogenous to gender norms. In fact, we would expect that parents and non-parents are different in many accounts, among them attitudes towards gender norms, e.g. women with more traditional views on the role of women might be more likely to become parents. If that is the case, then a direct comparison of parents and non-parents does not make much sense. Hence, in our analysis we restrict the sample to individuals who will eventually become parents. An alternative, but observationally equivalent, perspective is that any child affects views and behaviors, and we want to understand the differential effect of having a daughter.

Although the gender of the first child is exogenous, it can influence subsequent fertility decisions. Evidence from Germany indicates a slight preference for daughters; specifically, parents of a first-born son are marginally more likely to have an additional child. In the context of this study, such a daughter preference implies that we cannot condition on the number of children within a given period. For example, comparing parents with only one daughter (and no son) to those with only one son (and no daughter) may not yield an unbiased estimate of the effect of having a daughter on gender attitudes, as parents may self-select out of these groups according to their fertility stopping rules (e.g., a preference for sons), which may themselves be correlated with differences in gender norms.

This inability to condition on the number of children—due to self-selection out of the treatment group—implies that we test a slight variation of hypotheses H1 and H4. Specifically, we examine whether having a first-born daughter leads respondents to express different attitudes toward gender norms compared to having a first-born son. Since the gender of the first child is randomly assigned, this estimate can be interpreted causally. While a simple t-test could, in principle, be used,

we include year-of-birth dummy variables to restrict comparisons to parents who had their first child in the same year. For H4, we further interact the gender of the first child with a dummy for the respondent's gender, which identifies the differential effect of having a daughter among men.
We refrain from including variables from the period after giving birth, as these variables might be bad predictors, either in the sense of Angrist and Pischke, where accounting for these variables reintroduces selection errors; or as in Cinelli et al (2022), where these additional variables are channels through which having a daughter affects the outcome variables.

Hypotheses 2 and 3:

The data for testing hypotheses H2 and H3 are richer, allowing for more complex research designs. In this case, we exploit longitudinal data and implement a difference-in-differences (DiD) approach. An individual (H2) or household (H3) is considered treated from the moment the first daughter is born. We expect the treatment effects to be dynamic: as daughters grow older, their influence on parents' political orientations and gender attitudes may strengthen.

Two key considerations guide our design:

* Dynamic Effects. We expect parental attitudes and behaviors to evolve gradually as parents observe their daughters develop and interact with the world. To account for these dynamic effects, we employ the extended two-way fixed effects (TWFE) difference-in-differences estimator proposed by Wooldridge (2021), which allows for the estimation of heterogeneous treatment effects by treatment cohort and calendar year.

* First-Daughter Effects. Both theoretical and methodological considerations lead us to focus on the first daughter. Theoretically, subsequent daughters provide little new information, as parents are likely to have already adjusted their views after the first daughter. Methodologically, parents with a first-born daughter are already "treated," and additional daughters would represent a change in treatment intensity, but this change in treatment intensity is endogenous.

Since we use a difference-in-differences approach, we also rely on the no-anticipation assumption and some version of parallel trends. The no anticipation assumption can be controversial. Parents to be can change the distribution of household chores before the birth of a child, as women might not be able (or willing) to perform certain chores during pregnancy. However, for it to affect the estimates it must be the case that anticipation effects are different depending on the gender of a child.

Implementation
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Hypotheses 1 and 4:

The 2012 survey contains several variables that reflect gender norms (e.g. whether children suffers when one parent works). Instead of evaluating variables one by one, we use PCA to reduce the dimensionality

to the first eigenvalue for the main regression. In this new variable higher values indicate tendency to greater gender equality.

The following variables were used in the PCA. All data was collected in 2012

* plh0298_v1: Child under 6 suffers if mother works (FID) [2012]
* plh0302_v1: Child under 3 suffers if mother works (FID) [2012]
* plh0308_v1: Family: Man and woman should share equally in work and household/family duties
* plh0301: Women should focus more on the family than on their career

We test these hypotheses on a specific subsample of individuals. They have to meet some criteria:
- they were parents at the time of the survey. We drop those observations for whom (kidgeb01<0 | kidgeb01>2012).
- they provide information on the gender of their first born (kidsex01 >0)
- they answer all questions on gender attitudes.
- they live in private households headed by a German national. We identify these as individuals for whom the variable pop equals one or two.

We also use information on gender of the respondent and date of birth of the first child as additional controls in the regression analysis. In the regression, we use Heteroskedasticity Consistent standard errors implemented as "vce(hc3)" . These standard errors are more conservative, and once we introduced year of birth fixed effects they could not be estimated in the presence of singletons, individuals with a unique year of birth. Hence, in the final regression we also excluded these individuals. In the 5% sample it was a small amount, in a larger sample we expect them to be even fewer.

Hypothesis 2:

We consider sympathy for a political party as an indicator of right/left political leaning. We create a binary variable that equals one if an individual shows support for a party on the left of the political spectrum. When mapping political parties into a single axes (left-right) we use the data from the Chapel Hill Expert Survey (https://www.chesdata.eu/ches-europe). The database contains expert assessments of where parties are located along many dimensions. We focus on the economic dimension to identify right and left wing parties.

The approach has two clear advantages. First, the question on political preferences is available every year. This allows clean before-after comparisons. Second, studying a binary variable is considerably simpler than using multinomial models that consider voting for each party. However, it also introduces the need to include period fixed effects, for example to account for which parties are in power.

We envision two limitations of our approach. First, we assume that political parties are persistent in their views, which need not be the case. Political parties can adopt more leftist or rightist posture in response to electorate demands. Fortunately, this concerns is minor. According to CHES data, parties changed their behaviors, but only within the two camps. In other words, a party on the right might adopt more

extreme views in a year, and more centrist in the following, but it will not cross to the left. Second, parties can vary in terms of economic orientation (traditional axis), but also in terms of social orientation / preferences for environment. It would not be strange to observe parties that are very liberal and conservative. We focus on the first dimension only.

We use the variable plh0012_v6 to identify political orientation. Individuals have more leftist views when they express sympathy for one of the following parties (labels taken from the variables).

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      [1] [de] SPD
      [5] [de] Buendnis90/Green
      [6] [de] Die Linke
      [9] [de] SPD/Buendnis90/Green
      [16] [de] Buendnis90/Green
      [17] [de] SPD/Die Linke
      [26] [de] Piratenpart
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Our sample includes only observations that meet the following criteria.
- They are or will eventually become parents. We identify these individuals using the condition kidpnr01 > 0 & !mi(kidpnr01)
- They are German nationals (pgnation==1)
- They live in private households
- They are not always treated; that is, we exclude individuals who already had a daughter at the time of their first interview. Moreover, to avoid a setup where the control group is on average older than the treated group, we also exclude individuals who already had a boy at the time of the first interview. Keeping individuals who enter our data with a boy in the control group would introduce an asymmetry in the average childbearing phase between the treatment and control groups.

The treatment variable equals one on the date the first daughter is born, and remains equal to one for all subsequent years. We use the variables: syear, kidsex??, and kidgeb?? to create the treatment variable.

For our estimates to be interpreted as causal, there should be no evidence of differences in pre-trends. Of course, the absence of pre-trends does not guarantee parallel trends in subsequent periods, but it makes them more plausible. Our code includes additional tests for differences in pre-trends, which estimate the treatment effects in the treated groups but before their actual treatment. These pseudo-treatment effects are calculated using the never-treated group as a control. If this test passes and the pseudo-treatment effects (those before treatment) are not significant, then, for the sake of statistical power, we include the not-yet-treated units as controls in our final model.

Hypothesis 3:

Hypothesis 3 is different from others in that the unit of analysis is the household and not individual respondents.

SOEP includes several variables that measure activities thorough the day. As in H1 and H4, we do not consider these variables in isolation. Instead we build the Duncan segregation index for tasks to measure the extent of gender differentiation of tasks within the household. Higher values of this index would suggest greater inequality. While it is used mostly in

the context of labor markets, we believe the application here is appropriate.

The list of tasks considered in the analysis are:
* pli0038_h - Job, apprenticeship, second job (including travel time to and from work)
* pli0040 - Errands (shopping, trips to government agencies, etc.)
* pli0043_v3 - Housework (washing, cooking, cleaning)
* pli0044_v3 - Childcare
* pli0046 - Care and support of persons in need of care
* pli0047_v1 - Education or further training (also school, university)
* pli0049_v3 - Repairs on and around the house, car repairs, garden work
* pli0051 - Hobbies and other free-time activities

For the construction of the dependent variable (segregation index) at the household level, we first sum up the time spent by each household member in each activity (total_time = sum(activity_vars)) . Then, we obtain the share of time in each activity (activity_i / total_time). Later, we obtain the absolute value of differences between partner for each share, and add up these absolute differences.

The index has one important limitation. It considers the share of tasks. It could also happen that there are differences in total_time within couples. These differences will not be capture by the index.

Our sample includes only observations that meet the following criteria.
- They are or will eventually become parents. We identify these individuals using the condition kidpnr01 > 0 & !mi(kidpnr01)
- They are German nationals (pgnation==1)
- They live in private households
- They live in heterosexual couples
- They are not in patchwork families (e.g. if a partner has a daughter/son from a different marriage, they are excluded).
- They have performed at least one task at the time of the survey.
- If more than one couple live in a single address, we keep the younger couple.
- They are not always treated; that is, we exclude couples who already had a daughter at the time of their first interview. Moreover, to avoid a setup where the control group is on average older than the treated group, we also exclude couples who already had a boy at the time of the first interview. Keeping couples who enter our data with a boy in the control group would introduce an asymmetry in the average childbearing phase between the treatment and control groups.

We identify couples using the variable parid. Having matched individuals with the variables parid and id, we apply the criteria stated above.
-- gender1 != gender 2
-- the is no difference in the person id of the first five children (to exclude patchwork families).
-- We identify household with more than one couple using the household id (HID), and exclude the older couple using the age of members.

The treatment variable equals one on the date the first daughter is born, and remains equal to one for all subsequent years. We use the variables: syear, kidsex??, and kidgeb?? to create the treatment variable.

For our estimates to be interpreted as causal, there should be no evidence of differences in pre-trends. Of course, the absence of pre-trends does not guarantee parallel trends in subsequent periods, but it makes them more plausible. Our code includes additional tests for differences in pre-trends, which estimate the treatment effects in the treated groups but before their actual treatment. These pseudo-treatment effects are calculated using the never-treated group as a control. If this test passes and the pseudo-treatment effects (those before treatment) are not significant, then, for the sake of statistical power, we include the not-yet-treated units jointly with the never-treated units as controls in our final model.

References

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